

Chapter 6: Placental Abnormalities — Revision Notes

Williams Obstetrics 26e, pp. 295–329 of the PDF. High-yield numbers are in **bold**. The two tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 7 figures are included from the PDF.

Big picture: Inspect every placenta and cord in the delivery room; send for pathology based on clinical and placental findings. Most shape variants and small lesions are harmless, but some (succenturiate lobe, velamentous insertion, vasa previa, large chorioangioma, maternal floor infarction, cord thrombosis) threaten the fetus.

1. Placental Examination

- **Visual inspection by the obstetrician is recommended; routine pathology is not mandatory.** At minimum inspect the placenta and cord in the delivery room.
- CAP lists many indications, but data don't support all of them.

Table 6-1. Some Indications for Placental Pathological Examination^a

Maternal Indications	Fetal and Neonatal Indications	Placental Indications
Abruption	Admission to an acute care nursery	Gross lesions
Antepartum infection with fetal risks	Birth weight <10th or >95th percentile	Markedly abnormal placental shape or size
Anti-CDE alloimmunization	Fetal anemia	Markedly adhered placenta
Cesarean hysterectomy	Fetal or neonatal compromise	Term cord <32 cm or >100 cm*
Oligohydramnios or hydramnios	Neonatal seizures	Umbilical cord lesions
Peripartum fever or infection	Hydrops fetalis	Velamentous cord insertion
Preterm (<32 wk) delivery	Infection or sepsis	
Postterm (>42 wk) delivery	Major anomalies or abnormal karyotype	
Severe trauma	Multifetal gestation	
Suspected placental injury	Stillbirth or neonatal death	
Systemic disorders with known placental effects	Vanishing twin beyond the first trimester	
Thick meconium		
Unexplained late pregnancy bleeding		
Unexplained or recurrent pregnancy complications		

^aIndications are organized alphabetically. *The printed table reads "Term cord >32 cm or <100 cm", an obvious typo (that would include every normal cord). The intended criterion is a short (<32 cm) or long (>100 cm) cord, matching the text's normal 40–70 cm range.

2. Normal Placenta

Feature	Normal at term
Weight	470 g
Shape / diameter	Round to oval, 22 cm
Central thickness	2.5 cm (sonographically 2–4 cm , homogeneous)
Cord	Three vessels , usually central insertion
Retroplacental space	Hypoechoic, <1–2 cm

- **Basal plate** (maternal, against uterus) divided by clefts into **cotyledons**; **chorionic plate** (fetal) where the cord inserts.
- On the chorionic plate, **fetal arteries almost always cross over veins**. The amnion covering it peels off easily.
- **AIUM**: record **placental location and its relation to the internal os**; image the cord insertions and **count cord vessels** at every prenatal scan.
- Placenta accreta spectrum, previa (Ch 41) and GTD (Ch 20) are covered elsewhere.

3. Shape and Size Variants

Variant	Description	Clinical significance
Bilobate (bipartite, placenta duplex)	Two nearly equal discs; cord inserts between them (chorionic bridge or membranes)	Type 2 vasa previa risk
Multilobate	≥3 equal lobes	Rare
Succenturiate lobe	≥1 smaller accessory lobe in the membranes away from the main disc; vessels run in the membranes	Vasa previa if vessels overlie the cervix. Retained lobe → postpartum atony, hemorrhage or endometritis
Placenta membranacea	Villi cover all or nearly all of the cavity	Serious hemorrhage (associated previa/accreta)
Ring-shaped (annular)	Partial or complete ring; may be a variant of membranacea	↑ ante- and postpartum bleeding, FGR
Placenta fenestrata	Central part of disc missing (usually villi only; chorionic plate intact)	—

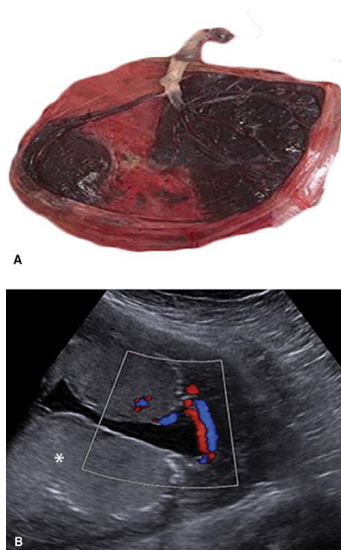


Figure 6-1. Succenturiate lobe: vessels run through the membranes from the main disc to the accessory lobe (A); color Doppler shows them crossing the cavity (B).

Placentomegaly

- Thickness increases **~1 mm/wk**; normally **≤40 mm**. **Placentomegaly = >40 mm**.

Pattern	Causes
Homogeneous thickening (villous enlargement)	Maternal: diabetes, severe anemia. Fetal: hydrops, anemia, syphilis , toxoplasmosis , parvovirus , CMV
Inhomogeneous (cystic)	Partial mole (edematous villi = small anechoic cysts); placental mesenchymal dysplasia (enlarged stem villi, diploid , no excess trophoblast proliferation)
Inhomogeneous (blood/fibrin)	Massive perivillous fibrin, intervillous or subchorionic thromboses, large retroplacental hematomas

4. Extrachorial Placentation

- **Chorionic plate is smaller than the basal plate** (doesn't reach the disc edge).

	Circummarginate	Circumvallate
Edge	Fibrin and old hemorrhage between disc and thin single-layer amniochorion	Thickened, opaque, gray-white ridge = double fold of chorion + amnion
Sonography	—	Thick linear band edge to edge; on cross section two "shelves" above opposite placental margins

- Circumvallate diagnosed **postpartum** (small studies): ↑ antepartum bleeding, abruption, fetal demise, preterm birth. But prospectively (17 cases) most sonographic circumvallate placentas were **transient**, and persistent ones were **benign**.
- **Most uncomplicated pregnancies with either type have normal outcomes → no extra surveillance usually needed.**

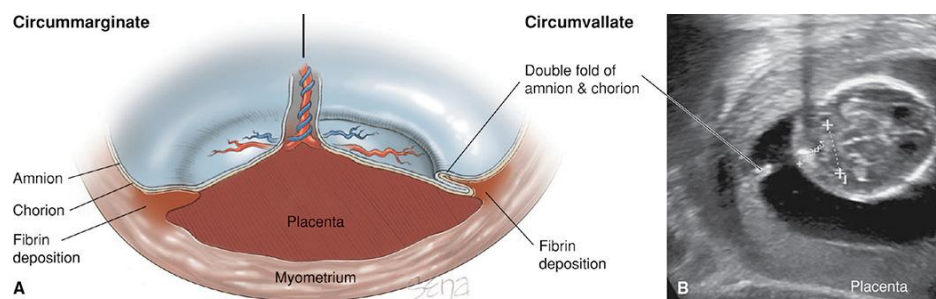


Figure 6-2. Circummarginate (single-layer amniochorion over fibrin) vs circumvallate (double fold of amnion and chorion forming a ridge); sonogram of the circumvallate shelf.

Table 6-2. Sonographic Bands During Pregnancy

Condition	Sonographic Findings
Normal early chorioamnionic separation	Crescent-shaped amnion mirrors the chorion's curve; distinct from the fetus; fuses after 16 weeks' gestation
Subchorionic hematoma	Echogenic blood lies between the myometrium and chorioamnion, which appears as a thin band crossing the cavity. Hemorrhage and band resolve over time
Uterine synechiae (amnionic sheet)	2.5- to 4.0-mm-thick, broad-based band crosses the cavity. Appears shelflike on cross section
Circumvallate placenta	Broad-based band extends from one placental edge to the other, just above the placental surface. Appears shelflike on cross section
Amnionic band	Thin strands cross and appear to tether fetal parts
Pseudoamnionic band syndrome	Thin strands tether fetal parts and form after fetoscopic surgeries or amniocenteses complicated by membrane laceration
Uterine septum	Chorioamnionic sac of an early pregnancy fills one horn of a septate or partial bicornuate uterus. Thick band of echoes, which may be wedge-shaped, extends from uterine fundus in midline
Membranes from vanishing twin	Depending on chorionicity, either a thin amnion or thicker chorioamnion spans the cavity
Placental vessels supported by membranes: velamentous insertion, succenturiate lobe	With grayscale imaging, vessels appear as bands. Color Doppler will clarify (see Figs. 6-1 and 6-6)

Modified from Dashe, 2017; Lafitte, 2017.

5. Circulatory Disturbances

- Two groups: **maternal blood flow** to/within the intervillous space disrupted, or **fetal blood flow** through villi disrupted.
- Common in normal mature placentas; functional reserve protects: **up to 30% of villi can be lost without fetal harm**. Extensive lesions → **FGR**.
- Many appear as focal **sonolucencies**. **Small isolated sonolucencies without complications = incidental**.

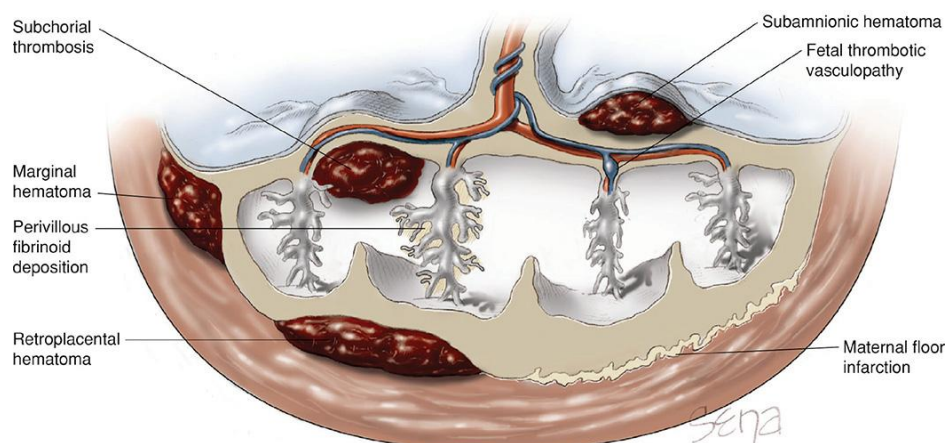


Figure 6-3. Sites of circulatory lesions: subchorial thrombus, subamniotic, marginal and retroplacental hematomas, perivillous fibrin, fetal thrombotic vasculopathy and maternal floor infarction.

Maternal blood flow disruption

Lesion	Mechanism / appearance	Significance
Subchorionic fibrin	Stasis near the chorionic plate → white/yellow firm raised plaques under the chorionic plate	Usually benign
Perivillous fibrin	Stasis around a villus → ↓ oxygenation, syncytial necrosis; small yellow-white nodules	Normal aging within limits. >25% of villi → FGR and adverse neonatal outcomes
Maternal floor infarction (massive perivillous fibrin if it extends up to entrap villi)	Dense fibrinoid layer in the basal plate , thick yellow-white corrugated surface; not a true infarct . Auto-/alloimmunity, APS , preeclampsia angiogenic factors	Miscarriage, FGR, preterm delivery, stillbirth; can recur . Not reliably seen on sonography (may thicken the basal plate)
Intervillous thrombus	Maternal blood mixed with fetal blood from a villous break; red (new) or white-yellow (old), up to several cm	Common, usually harmless. A maternal-fetal communication → large ones raise MSAFP
Infarction	Villi get O ₂ only from maternal intervillous blood; obstructed supply → infarct	Few are benign; many → placental insufficiency . Thick, central, random infarcts → preeclampsia or lupus anticoagulant

Hematomas

Type	Location
Retroplacental	Between placenta and decidua (abruption = a large, clinically significant retroplacental hematoma)
Marginal (= clinical subchorionic hemorrhage)	Between chorion and decidua at the placental edge
Subamniotic	Fetal vessel origin; beneath amnion, above chorionic plate
Subchorial thrombus	Roof of the intervillous space, under the chorionic plate. Massive = Breus mole

Sonographic evolution of a hematoma

Time after bleed	Echogenicity
1st week	Hyperechoic to isoechoic
1–2 weeks	Hypoechoic
>2 weeks	Anechoic

- Most visible subchorionic hematomas are small and harmless. **Extensive** retroplacental, marginal or subchorial collections → ↑ **miscarriage, stillbirth, abruption, preterm delivery**.

Fetal blood flow disruption

- **Fetal vascular malperfusion**: thrombosis in fetal vessels makes distal villi nonfunctional. A few thrombi are normal; many (e.g., **preeclampsia**) → **FGR, stillbirth, nonreassuring FHR**.
- **Villous vascular lesions**:

Lesion	Definition	Associations
Chorangiosis	↑ capillaries in terminal villi: ≥10 capillaries in ≥10 villi in ≥10 fields at 10× ("rule of 10s")	Long-standing hypoperfusion/hypoxia. Focal form → lower Apgars, fetal vascular malperfusion
Chorangiomas	↑ capillaries in stem villi; terminal villi spared	FGR, anomalies

- Clinical significance of both remains unclear.
- **Subamniotic hematoma**: usually **acute, iatrogenic, harmless** (cord traction in 3rd stage ruptures a vessel near the insertion). Large chronic antepartum ones → fetomaternal hemorrhage or FGR; can mimic chorioangioma → **color Doppler shows no internal flow**.

6. Placental Calcification

- Most common on the **basal plate**; ↑ with gestation, **smoking** and higher maternal serum calcium.

Grannum grading (0–3)

Grade	Sonographic appearance
0	Homogeneous, no calcification, smooth flat chorionic plate
1	Scattered echogenicities; subtle chorionic plate undulations
2	Basal plate stippling; comma-shaped densities from an indented chorionic plate that fall short of the basal plate
3	Indentations run from chorionic to basal plate → cotyledon-like compartments; ↑ basal densities

- **Poor predictor of neonatal outcome near term.** But **grade 3 before 32 wk** has been linked to **stillbirth** and other adverse outcomes (two small studies).

7. Placental Tumors

Chorioangioma (chorangioma)

- Benign tumor of villous vessels and stroma; incidence **~1%**.
- Fetal-to-maternal bleeding across tumor capillaries → ↑ **MSAFP** → sonography to exclude a neural-tube defect.
- **Sonography:** well-circumscribed, rounded, mostly **hypoechoic** mass near the **chorionic plate** protruding into the amniotic cavity. ↑ **color Doppler flow** distinguishes it from hematoma, partial mole, teratoma, metastases, leiomyoma. (Rare chorangiocarcinoma mimics it.)

Size	Consequences
Small	Usually asymptomatic
Large (>4 cm)	AV shunting → high-output heart failure, hydrops, fetal death; microangiopathic hemolytic anemia; hydramnios, preterm delivery, FGR

- **Large size and hydrops** are the main predictors of poor outcome.
- **Work-up:** grayscale + color Doppler, amniotic fluid volume; **MSAFP and Kleihauer-Betke** for fetomaternal hemorrhage; **fetal echo** (cardiac function); **MCA Doppler** (anemia).
- **Treatment (specialized centers): endoscopic laser ablation of feeder vessels** (most used; favorable outcomes); fetal transfusion for anemia, amnioreduction for hydramnios, digoxin for heart failure.

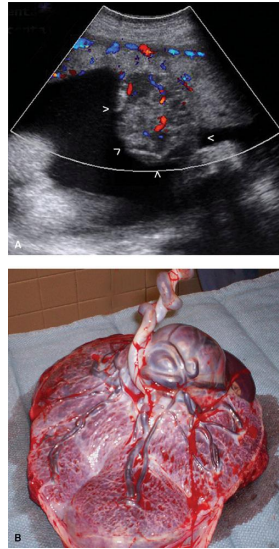


Figure 6-4. Chorioangioma: color Doppler shows internal flow in a well-circumscribed mass (A); grossly a round mass protruding from the fetal surface (B).

Metastatic tumors

- Rare. Most common maternal sources: **melanoma, leukemia/lymphoma, breast cancer**. Usually confined to the intervillous space; **spread to the fetus is uncommon but most often with melanoma**.
- Fetal tumors metastasizing to the placenta: rare, mostly **neuroectodermal**.

8. Amniochorion

Chorioamnionitis

- Genital flora **ascend**, usually after **prolonged membrane rupture and during labor**.
- Sequence: chorion + decidua **over the internal os** → full-thickness membranes (**chorioamnionitis**) → amnionic fluid → chorionic plate and cord (**funisitis**).
- Usually **microscopic/occult**; a possible explanation for many unexplained cases of **ruptured membranes and preterm labor**. Gross infection: clouded membranes ± foul odor.

Other membrane abnormalities

Condition	Features	Significance / management
Amnion nodosum	Small light-tan nodules on the amnion over the chorionic plate (fetal squames + fibrin); scrape off	Sign of prolonged severe oligohydramnios
Amnionic band sequence	Disruption sequence: bands tether, constrict or amputate fetal parts; spontaneous or after fetal surgery	Limb-reduction defects, facial clefts, encephalocele ; cord compromise. Severe spine/ventral wall defects → think limb-body wall complex . Sonography usually finds the sequelae , not the bands → targeted scan. Management by severity; fetoscopic laser release in selected cases
Amnionic sheet	Normal amniochorion draped over a uterine synechia	Little risk; slightly ↑ PPROM and abruption

- **Clues to amnionic bands:** limb-reduction defect, **encephalocele in an atypical location**, an extremity with edema or positional deformity.

9. Umbilical Cord

Length

- Usually **40–70 cm**; very few **<30 cm or >100 cm**. Longer with ↑ **parity and BMI**.
- **Short cords**: congenital malformations, intrapartum distress.
- **Long cords**: **entanglement, prolapse**, fetal anomalies.
- Cord diameter: lean cords ↔ poor growth, thick ↔ macrosomia (utility unclear).

Coiling

- Usually **sinistral (left-twisting)**. **Umbilical coiling index (UCI)** = complete coils per cm.
 - Normal UCI: **0.4 antepartum (sonographic)** vs **0.2 postpartum (measured)**.
 - **Hypocoiled <10th percentile; hypercoiled >90th percentile**.
- Significance controversial; extremes most consistently linked to **intrapartum FHR abnormalities, preterm labor, FGR**. Not part of standard sonography.

Vessel number

- Embryos have two umbilical veins; the **right vein atrophies in the 1st trimester** → 1 vein + 2 thick-walled arteries.
- **Four-vessel cord**: rare, often with anomalies; good prognosis if isolated.

Single umbilical artery (SUA): the most common aberration

Population	Incidence
Liveborn neonates	0.63%
Perinatal deaths	1.92%
Twins	3%

- Frequent with major malformations, mostly **cardiovascular and genitourinary** → **targeted sonography ± fetal echocardiography**.
- **SUA + another anomaly** → **aneuploidy risk greatly ↑** → **amniocentesis for karyotype**.
- **Isolated SUA** in a low-risk pregnancy: aneuploidy risk **not significantly ↑**; linked to FGR and perinatal death in some studies → **clinical growth monitoring is reasonable** (value of sonographic surveillance unclear).
- On sonography, the two arteries **encircle the fetal bladder** (extensions of the superior vesical arteries).
- **Fused umbilical artery** (shared lumen, usually near the placental insertion): ↑ marginal/velamentous insertion, not anomalies.
- **Hyrtil anastomosis**: connection between the two arteries near the placental insertion (in most placentas) → **equalizes pressure**, improves perfusion during contractions or compression of one artery. **SUA fetuses lack this safety valve**.

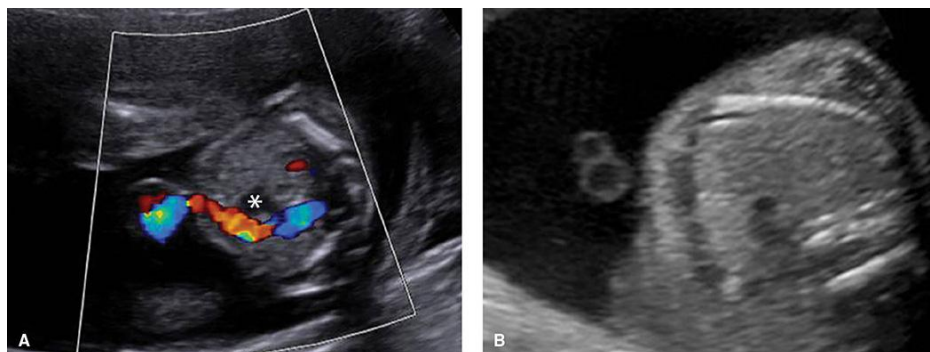


Figure 6-5. Single umbilical artery: only one artery runs beside the fetal bladder (A); cross section shows two vessels, one small artery and one larger vein (B).

Remnants and cysts

- **Allantoic duct, vitelline duct and embryonic vessel remnants** in **25–50%** of sectioned cords: not associated with malformations.

	True cysts	Pseudocysts
Origin	Epithelium-lined allantoic or vitelline remnants	Local degeneration of Wharton jelly
Site	Closer to the fetal insertion	Anywhere
Frequency	Less common	More common

- **Single 1st-trimester cyst** → **usually resolves**. **Multiple cysts** → **miscarriage or aneuploidy**. **Persistent beyond the 1st trimester** → **structural and chromosomal anomalies**.

Insertion

Insertion	Incidence	Significance
Central	Normal	—
Eccentric	Common	No identifiable risk
Marginal ("battledore")	6% singletons, 11% twins	Rarely problematic; cord may avulse at placental delivery; weight discordance in monochorionic twins
Velamentous	~1%; 6% in twins ; more common with placenta previa	Vessels run unprotected in the membranes → compression → hypoperfusion, acidemia. ↑ low Apgar, stillbirth, preterm delivery, SGA . Cord avulsion. Monitor growth (clinically or by sonography)
Furcate	Rare	Vessels lose Wharton jelly before inserting (amnion sheath only) → compression, twisting, thrombosis

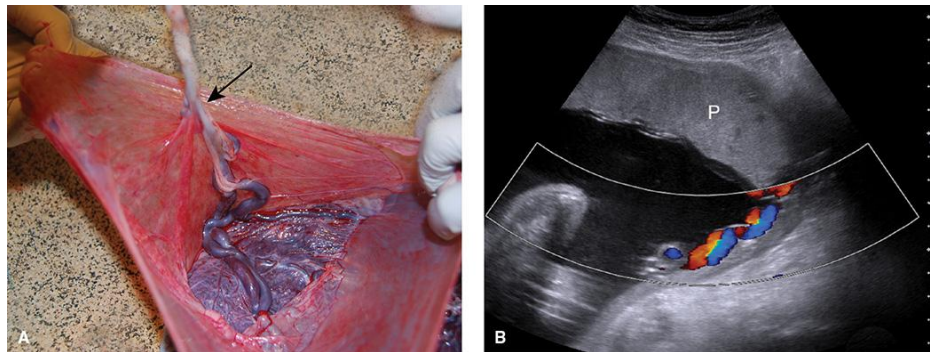


Figure 6-6. Velamentous insertion: the cord inserts into the membranes (arrow) and vessels travel unprotected to the disc (A); color Doppler shows them running along the uterine wall (B).

Vasa previa

- Fetal vessels **in the membranes over the cervical os** → tear with cervical dilation or membrane rupture → **rapid fetal exsanguination**; can also be compressed by the presenting part.
- Incidence **1 in 338–365** pregnancies.

Type	Vessels
Type 1	Part of a velamentous cord insertion
Type 2	Span between lobes of a bilobate or succenturiate placenta

- **Other risk factors: IVF conception; 2nd-trimester placenta previa** or low-lying placenta (even if it later migrates).
- **Antepartum diagnosis** → **perinatal survival 97–100%** (far better than intrapartum diagnosis).
- **Screening:** at the midtrimester scan; add **transvaginal sonography** in suspicious cases (vessels inserting into membranes and running over the internal os). **Color Doppler of the cord insertion**, especially with previa/low-

lying placenta; the waveform shows the **fetal heart rate** (umbilical artery pattern). Median prenatal detection **93%**.

Management

1. **Repeat imaging**: up to **39%** resolve.
2. **Bed rest**: no added benefit.
3. **Antenatal corticosteroids** as indicated or **prophylactically at 28–32 wk**.
4. **Consider hospitalization at 30–34 wk** (surveillance; rapid delivery for labor, bleeding or ROM), especially with risk factors for early delivery.
5. **ACOG: scheduled cesarean at 34–36 wk**. Deliver the fetus **immediately after hysterotomy**; **delayed cord clamping not encouraged**.
 - Fetoscopic laser ablation described in a few cases.
 - **Any unexplained ante- or intrapartum vaginal bleeding** → **consider vasa previa with a torn fetal vessel**. Often rapidly fatal. With less bleeding, **fetal vs maternal blood** can be told apart because **fetal hemoglobin resists alkali/acid denaturation**.

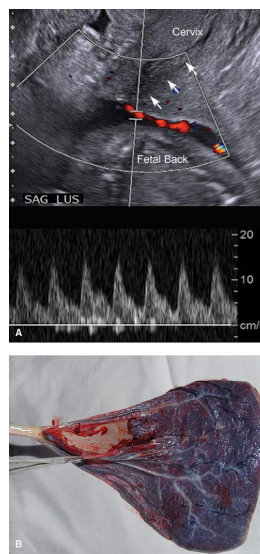


Figure 6-7. Vasa previa: color Doppler shows a fetal vessel over the internal os with an umbilical-artery waveform (A); amniotomy site through membranes carrying fetal vessels (B).

Knots, strictures and loops

Lesion	Frequency	Key facts
True knot	~1% of births	From fetal movement; risk factors hydramnios, diabetes ; very common/dangerous in monoamniotic twins . Singleton stillbirth ↑ 4–10× . Sonography: " hanging noose " sign; 3-D and color Doppler help. Vaginal delivery suitable ; FHR, cesarean rate and cord gases usually normal. Surveillance unclear (UA Doppler, NST, kick counts)
False knot	—	Focal vessel redundancy/folding; no significance
Cord stricture	Rare	Focal narrowing near the fetal insertion , absent Wharton jelly, obliterated vessels → usually stillbirth . Rarely from an amniotic band
Nuchal cord	1 loop 20–34% ; 2 loops 2.5–5% ; 3 loops 0.2–0.5%	Up to 20% have moderate–severe variable decelerations in labor with lower UA pH; not relieved by amnioinfusion . Not linked to adverse outcome ; vaginal delivery suitable
Funic presentation	Uncommon	Cord is the presenting part; with malpresentation. Risk of prolapse → cesarean at term

Vascular lesions

Lesion	Key facts
Cord hematoma	Rare; usually venous rupture into Wharton jelly. Linked to abnormal cord length, aneurysm, trauma, entanglement, cord venipuncture, funisitis. Hypoechoic mass without flow. → stillbirth , abnormal FHR (normal outcomes reported)
Cord vessel thrombosis	Rare. 70% venous, 20% venous + arterial, 10% arterial . High rates of stillbirth, FGR, intrapartum distress → consider prompt delivery if viable
Umbilical vein varix	Intraamniotic or intraabdominal; cystic dilation contiguous with normal vein. May compress an artery, rupture or thrombose. ~1/5 have other anomalies ; isolated ones do well with surveillance
Umbilical artery aneurysm	Congenital wall thinning with little Wharton jelly, near the placental insertion . Associated with SUA, trisomy 18 , fluid extremes, FGR, stillbirth. >5 cm → high-output failure risk . Cyst with hyperechoic rim; low-velocity or turbulent nonpulsatile flow. Consider karyotype, surveillance, early delivery; some advise cesarean

Quick-Recall: Lesion → Clinical Consequence

Finding	Main risk	Action
Succenturiate / bilobate	Vasa previa (type 2); retained lobe → PPH, endometritis	Check membranes for vessels; inspect placenta
Placenta membranacea / annular	Hemorrhage (previa, accreta); FGR	—
Placentomegaly (>40 mm), homogeneous	Diabetes, anemia, hydrops, TORCH/parvovirus/CMV/syphilis	Look for cause
Circumvallate	Usually benign	No extra surveillance
Perivillous fibrin >25% / maternal floor infarction	FGR, stillbirth; recurs	Counsel for future pregnancies
Large intervillous thrombus	↑ MSAFP	—
Many central infarcts	Preeclampsia, lupus anticoagulant	—
Extensive retroplacental / marginal hematoma	Miscarriage, abruption, stillbirth, preterm birth	—
Chorioangioma >4 cm	Hydrops, heart failure, anemia, hydramnios	Echo, MCA Doppler; laser ablation
Amnion nodosum	Severe prolonged oligohydramnios	—
Amnionic bands	Limb amputation, clefts, atypical encephalocele	Targeted scan; selected fetoscopic release
SUA + anomaly	Aneuploidy	Amniocentesis
Isolated SUA	Possible FGR	Monitor growth
Velamentous insertion	Stillbirth, SGA, vasa previa	Monitor growth; look for vasa previa
Vasa previa	Fetal exsanguination	Steroids 28–32 wk, admit 30–34 wk, cesarean 34–36 wk
True knot	Stillbirth 4–10×	Vaginal delivery OK
Cord thrombosis / stricture	Stillbirth	Consider prompt delivery

Key Numbers to Memorize

- Placenta **470 g, 22 cm, 2.5 cm** thick (sonography **2–4 cm**); retroplacental space **<1–2 cm**
- Thickness ↑ **1 mm/wk**; **>40 mm = placentomegaly**
- Up to **30%** of villi can be lost without harm; perivillous fibrin **>25%** → FGR

- Hematoma echogenicity: hyper/iso **<1 wk**, hypo **1–2 wk**, anechoic **>2 wk**
- Chorangioma: **10 × 10 × 10 at 10x**
- Chorioangioma **~1%**; **>4 cm** → complications
- Amnionic sheet band **2.5–4.0 mm**; chorioamnion fuses after **16 wk**
- Cord **40–70 cm** (few <30 or >100); UCI **0.4** antepartum / **0.2** postpartum
- SUA **0.63%** live births, **1.92%** perinatal deaths, **3%** twins
- Remnants in **25–50%** of cords
- Marginal insertion **6%** (twins **11%**); velamentous **1%** (twins **6%**)
- Vasa previa **1/338–365**; detection **93%**; **39%** resolve; survival **97–100%** if diagnosed antenatally; steroids **28–32 wk**, admit **30–34 wk**, deliver **34–36 wk**
- True knot **1%**, stillbirth **4–10x**; nuchal cord 1 loop **20–34%**, 2 loops **2.5–5%**, 3 loops **0.2–0.5%**; variables in up to **20%**
- Cord thrombosis **70%** venous / **20%** both / **10%** arterial; varix anomalies **~1/6**; artery aneurysm **>5 cm** → heart failure risk